

Exp no :01

Date :

Evidence Acquisition Using Access Data FTK Imager

Description

Forensic Toolkit or FTK is a computer forensics software product made by AccessData. This is a Windows based commercial product. For forensic investigations, the same development team has created a free version of the commercial product with fewer functionalities. This FTK Imager tool is capable of both acquiring and analyzing computer forensic evidence.

The evidence FTK Imager can acquire can be split into two main parts. They are:

- Acquiring volatile memory
- Acquiring non-volatile memory (Hard disk)

There are two possible ways this tool can be used in forensics image acquisitions:

- Using FTK Imager portable version in a USB pen drive or HDD and opening it directly from the evidence machine. This option is most frequently used in live data acquisition where the evidence PC/laptop is switched on.
- Installing FTK Imager on the investigator's laptop.

In this case the source disk should be mounted into the investigator's laptop via write blocker. The write blocker prevents data being modified in the evidence source disk while providing read-only access to the investigator's laptop. This helps to maintain the integrity of the source disk.

Acquiring volatile memory using FTK Imager

The FTK Imager tool helps investigators to collect the complete volatile memory (RAM) of a computer. The following steps will show you how to do this.

Open FTK Imager and navigate to the volatile memory icon (capture memory).

Memory Capture

Destination path:
C:\Drivers\FTK_MAGER

Browse

Destination filename:
memdump.mem

☒ Include pagefile
pagefile.sys

☒ Create AD1 file
memcapture.ad1

Capture Memory

Cancel

Memory Progress

Destination: C:\Drivers\FTK_MAGER\memdump.mem

Status: Dumping RAM: 16GB/19GB [83%]

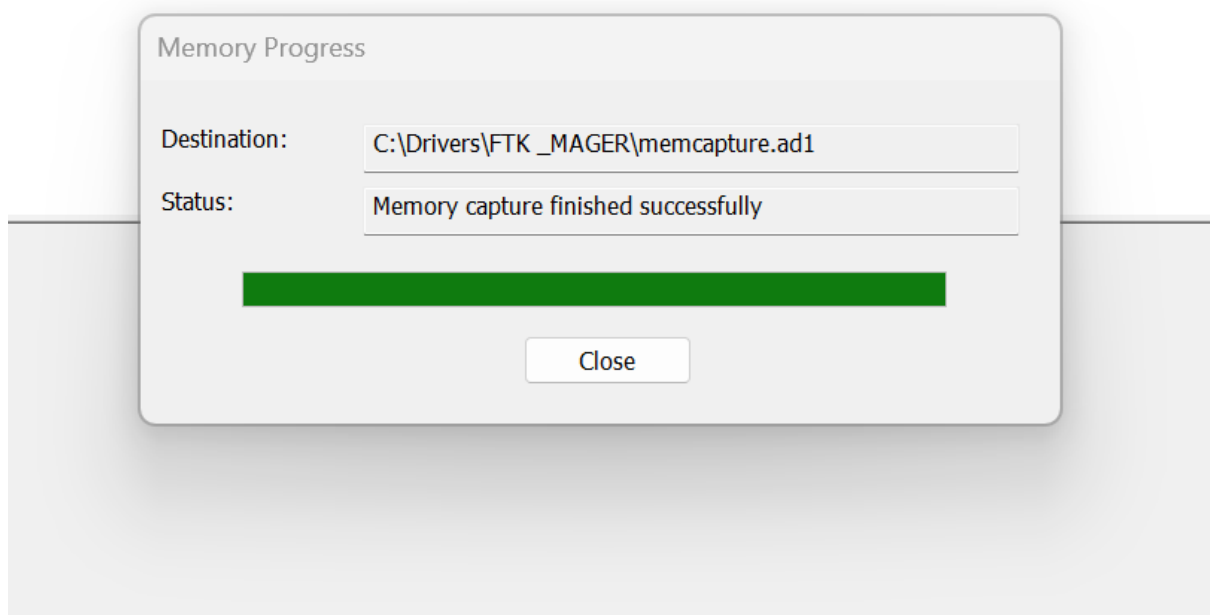
Cancel

NOTE: This tool provides options to include pagefile and AD1 files when acquiring the volatile memory.

Pagefile: The pagefile (pagefile.sys) is used in Windows operating systems as volatile memory due to limitation of physical random-access memory (RAM). It is located under the “C” partition ready to use as volatile memory when the existing RAM capacity is exceeded. So this file can have quite a bit of valuable data when considering the volatile memory. Therefore, it is recommended to capture and collect this file in the acquisition.

AD1 file: AD1 is the FTK imager image file. The investigator has the option to create an AD1 file for later use.

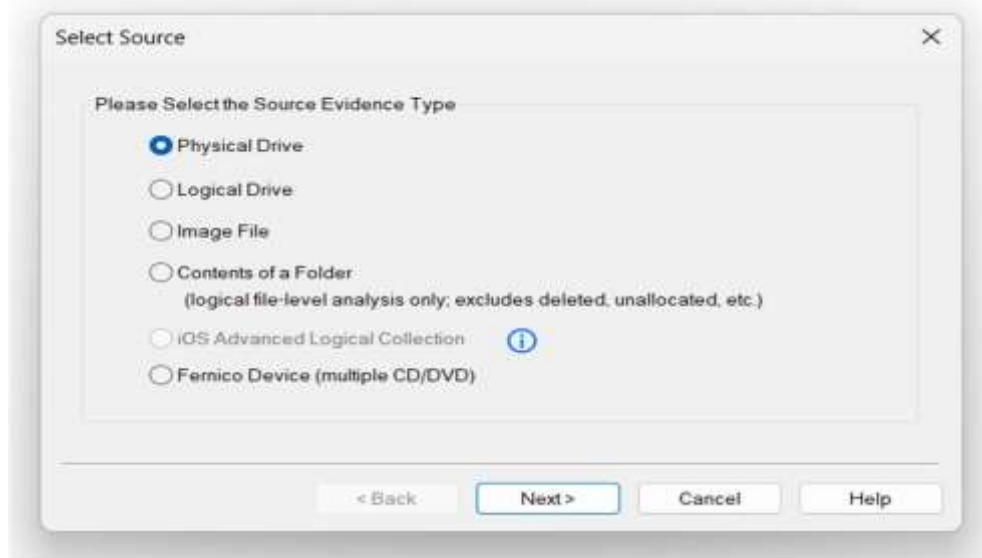
Clicking the “capture memory” button will start acquiring the volatile memory.



NOTE: Once the acquisition has completed, the destination folder will have the acquired memory with the file extension of “.mem”.

Acquiring non-volatile memory (Disk Image) using FTK Imager

As previously stated, this same tool can be used to collect a disk image as well. Open FTK Imager and navigate to “Create Disk Image”.



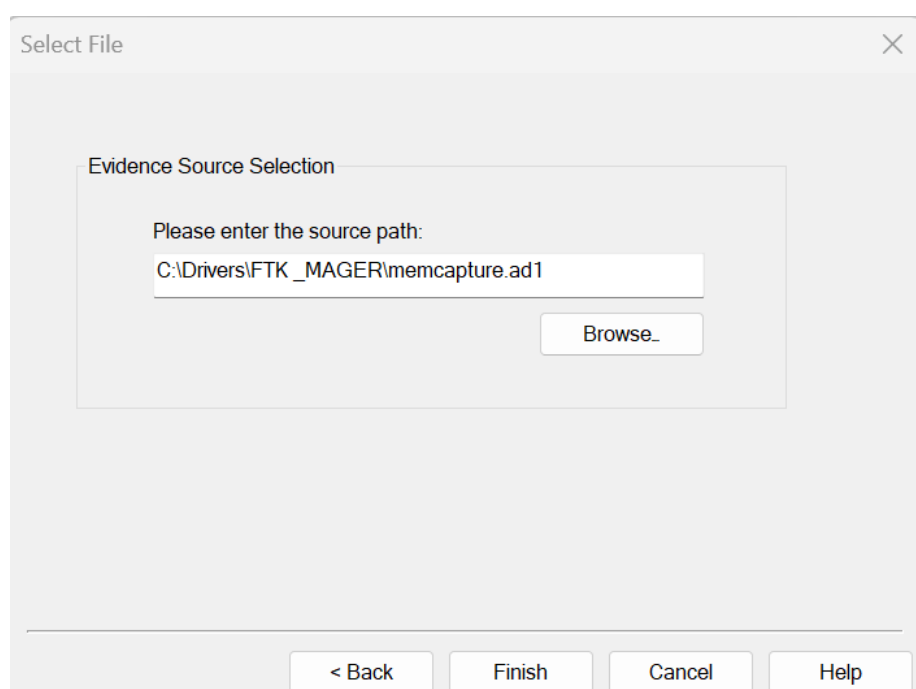
NOTE: FTK Imager is capable of acquiring physical drives (physical hard drives), logical drives (partitions), image files, contents of a folder, or CDs/DVDs. Investigators can connect external HDDs into the collection computer

via write blocker and use the “logical drive” option to select the mounted HDD as a partition.

Collecting Physical Drives

Select the “Physical Drive” option.

Select the drive you need to acquire and click “Finish”.



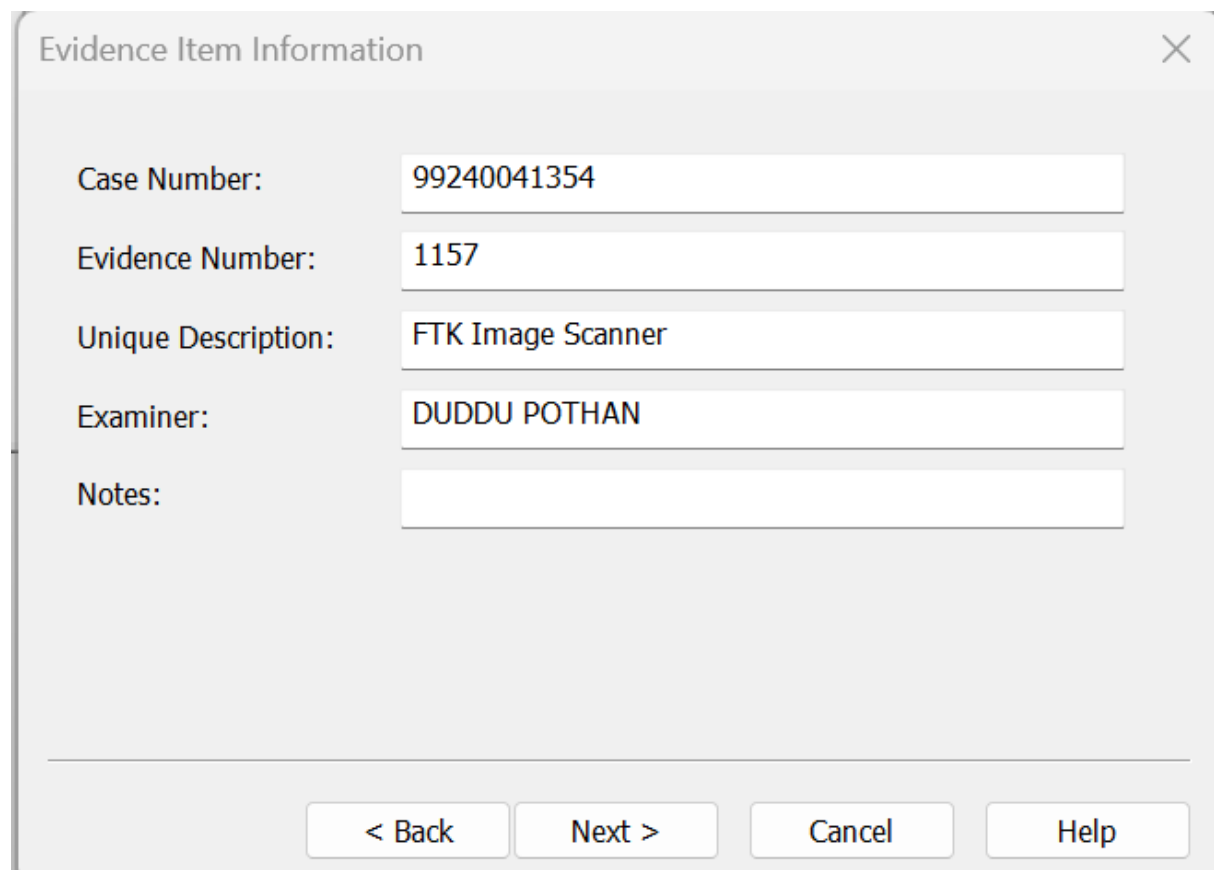
Raw (dd): This is the image format most commonly used by modern analysis tools. These raw file formatted images do not contain headers, metadata, or magic values. The raw format typically includes padding for any memory ranges that were intentionally skipped (i.e., device memory) or that could not be read by the acquisition tool, which helps maintain spatial integrity (relative offsets among data).

SMART: This file format is designed for Linux file systems. This format keeps the disk images as pure bitstreams with optional compression. The file consists of a standard 13-byte header followed by a series of sections. Each section includes its type string, a 64-bit offset to the next section, its 64-bit size, padding, and a CRC, in addition to actual data or comments, if applicable.

E01: this format is a proprietary format developed by Guidance Software's EnCase. This format compresses the image file. An image with this format starts with case information in the header and footer, which contains an MD5 hash of the entire bit stream. This case information contains the date and time of acquisition, examiner's name, special notes and an optional password.

AFF: Advance Forensic Format (AFF) was developed by Simson Garfinkel and Basis Technology. Its latest implementation is AFF4. The goal is to create a disk image format that does not lock the user into a proprietary format that may prevent them from being able to properly analyze it.

Now enter the case details.

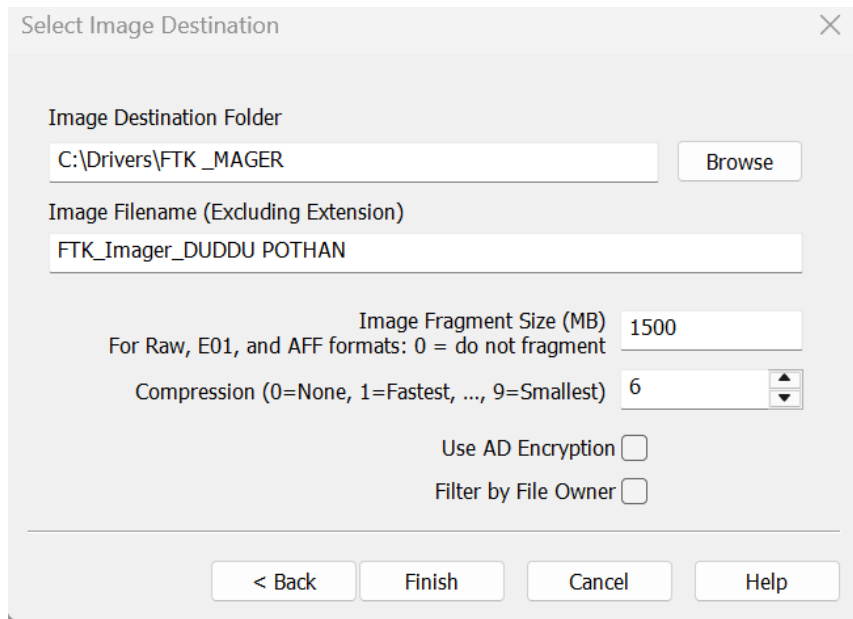


The image shows a software dialog box titled "Evidence Item Information" with a close button (X) in the top right corner. The dialog contains five input fields, each with a label to its left: "Case Number:" with the value "99240041354", "Evidence Number:" with the value "1157", "Unique Description:" with the value "FTK Image Scanner", "Examiner:" with the value "DUDDU POTHAN", and "Notes:" which is currently empty. At the bottom of the dialog, there are four buttons: "< Back", "Next >", "Cancel", and "Help".

Field	Value
Case Number:	99240041354
Evidence Number:	1157
Unique Description:	FTK Image Scanner
Examiner:	DUDDU POTHAN
Notes:	

< Back Next > Cancel Help

Add an image destination (where the image file will be saved), image file name and fragment size.



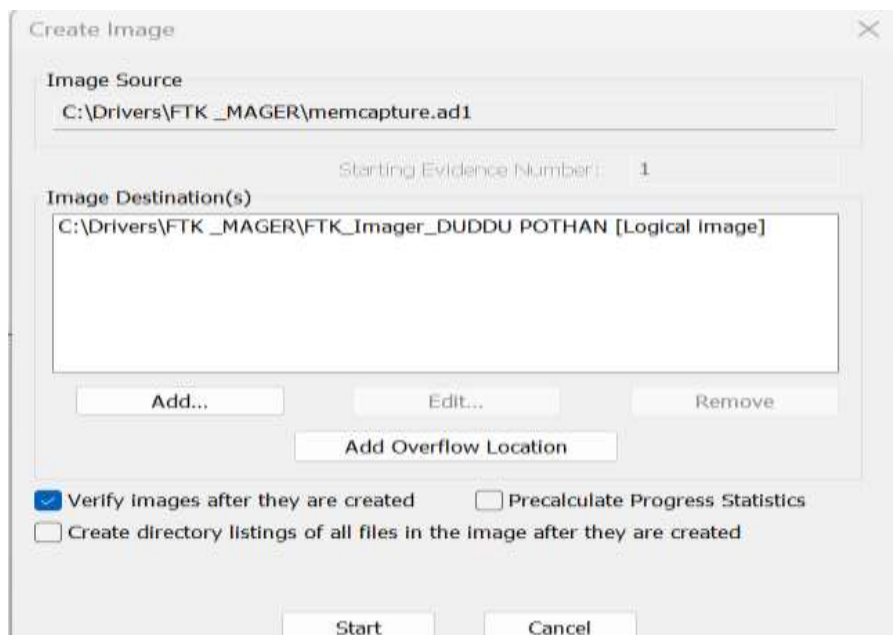
The "Select Image Destination" dialog box contains the following fields and controls:

- Image Destination Folder:** A text box containing "C:\Drivers\FTK_MAGER" and a "Browse" button.
- Image Filename (Excluding Extension):** A text box containing "FTK_Imager_DUDDU POTHAN".
- Image Fragment Size (MB):** A text box containing "1500". Below it, a note reads: "For Raw, E01, and AFF formats: 0 = do not fragment".
- Compression:** A dropdown menu showing "6". The label reads: "Compression (0=None, 1=Fastest, ..., 9=Smallest)".
- Use AD Encryption:** An unchecked checkbox.
- Filter by File Owner:** An unchecked checkbox.
- Buttons:** "< Back", "Finish", "Cancel", and "Help".

Image Fragment Size (MB): this option will separate the image file into multiple images and save them in the same destination. If you need only one file instead of creating multiple fragmented images, you must set the image fragment size to "0".

Select the "verify images after they are created" option. This will verify the hash values once the image has created. In order to ensure integrity, it is recommended to use this option. However, this will increase the time taken to acquire your evidence, especially if you're dealing with a large disk image size.

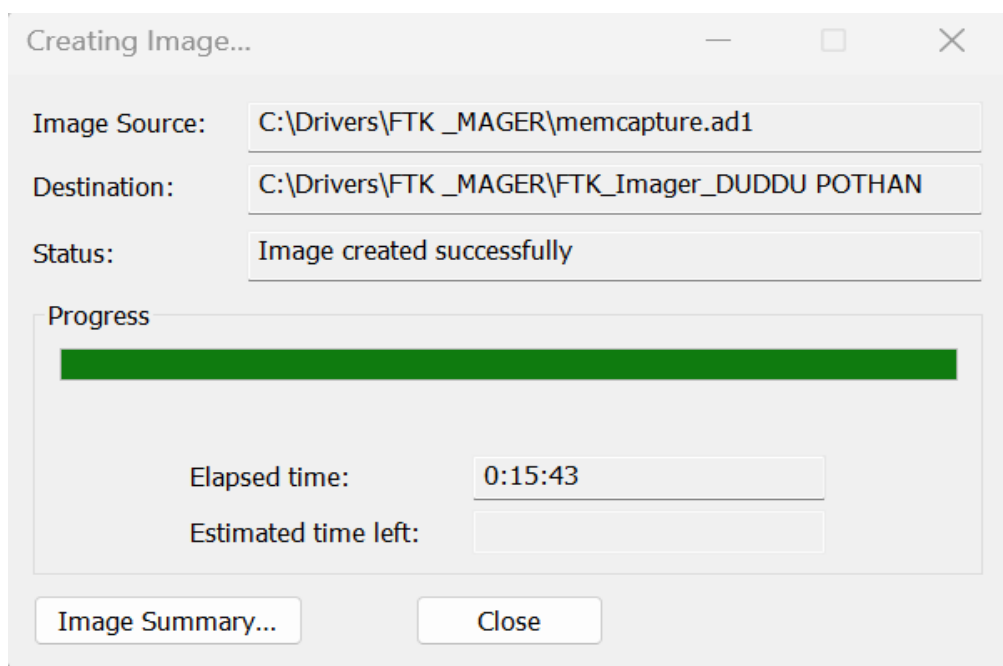
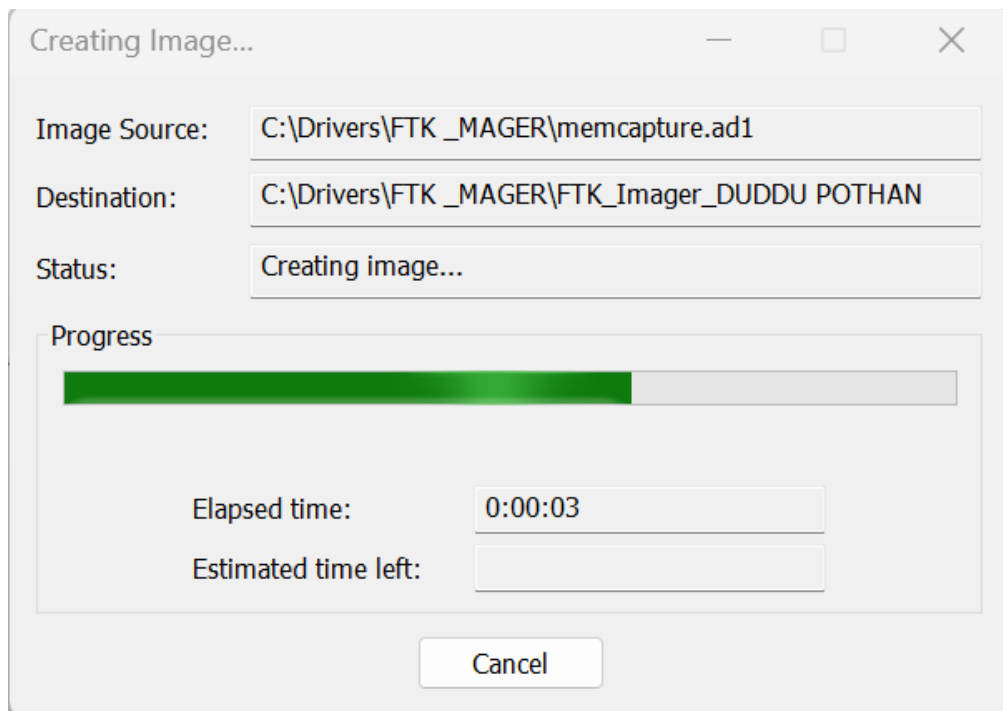
Click "start" to start acquiring.

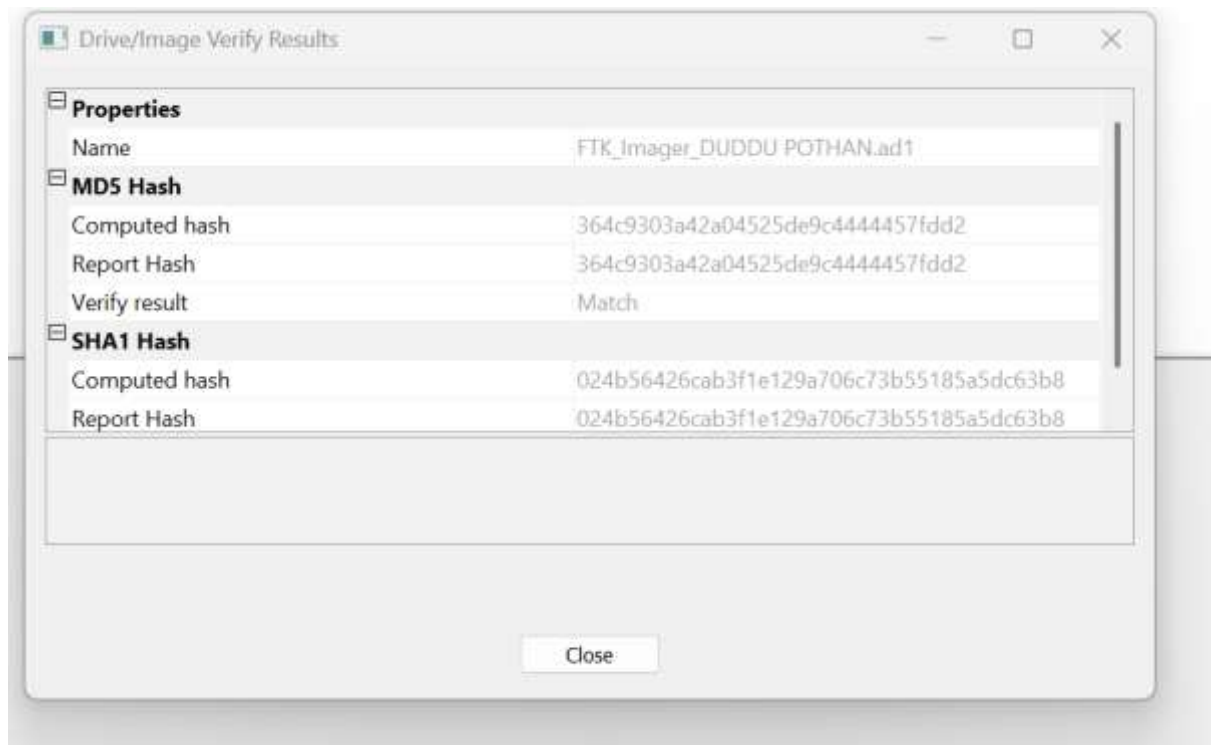


The "Create Image" dialog box contains the following fields and controls:

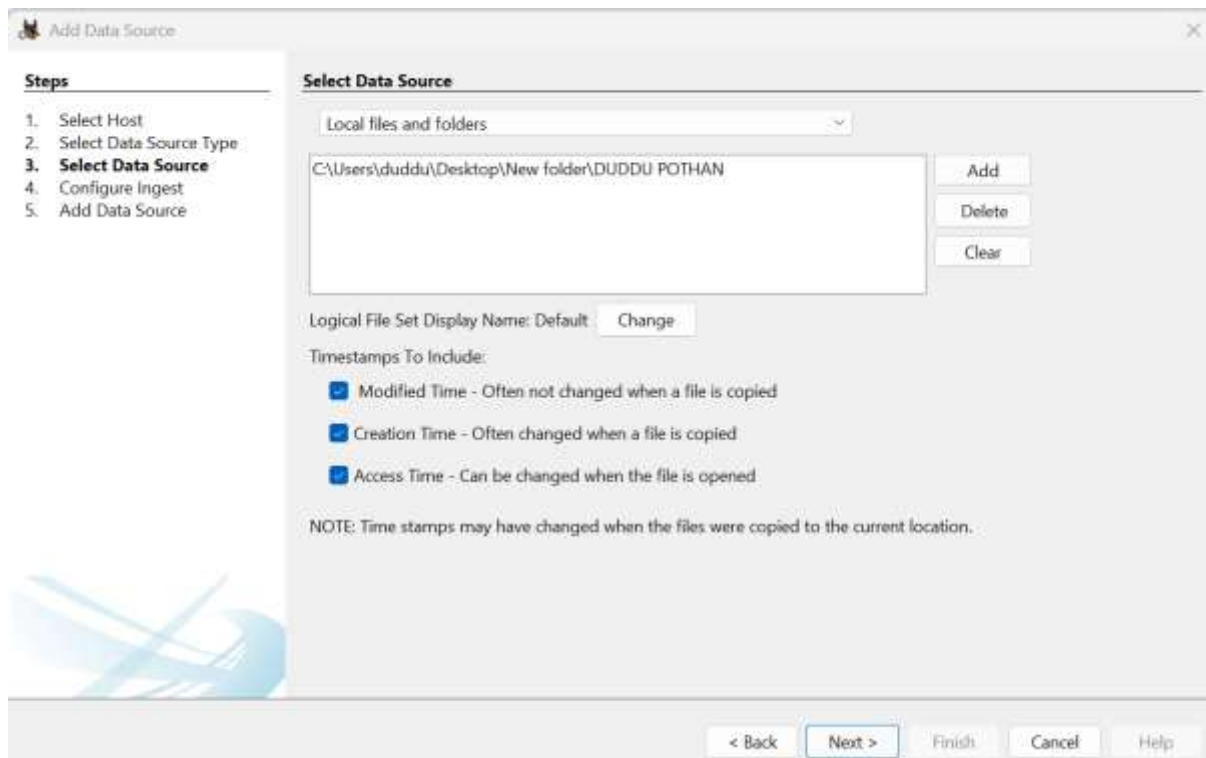
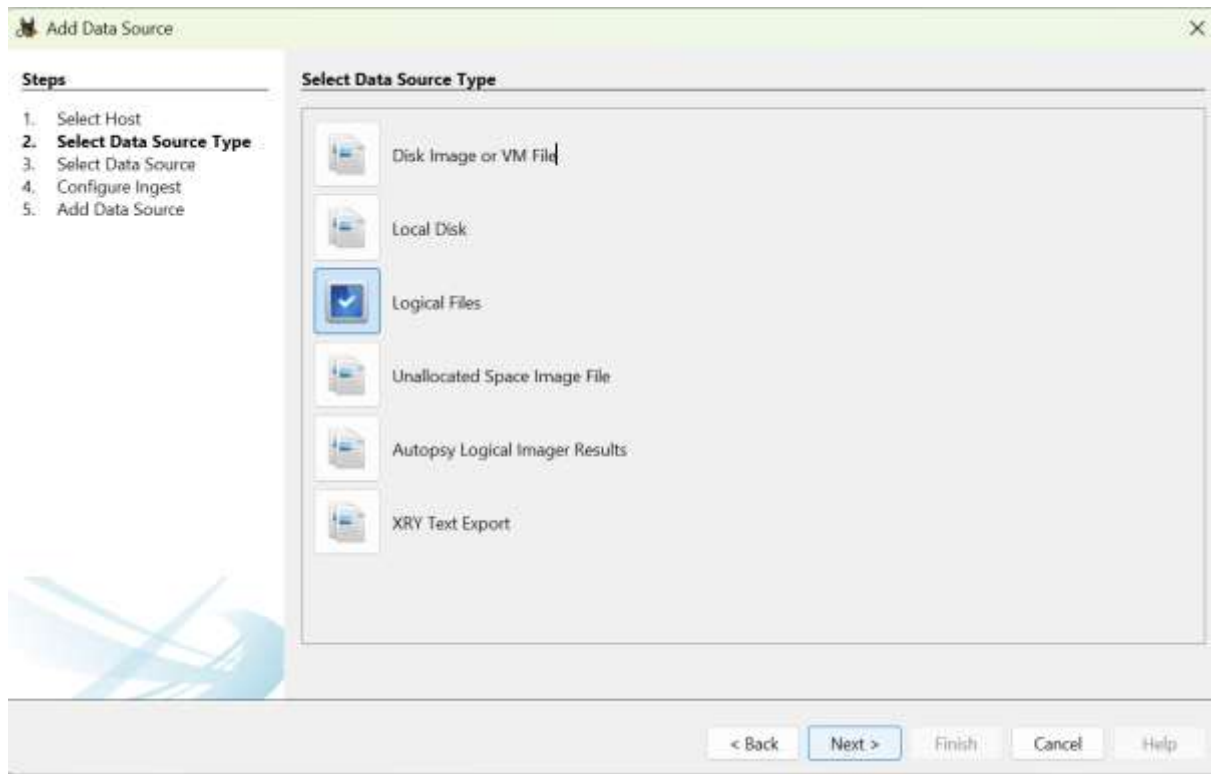
- Image Source:** A text box containing "C:\Drivers\FTK_MAGER\memcapture.ad1".
- Starting Evidence Number:** A text box containing "1".
- Image Destination(s):** A list box containing "C:\Drivers\FTK_MAGER\FTK_Imager_DUDDU POTHAN [Logical Image]".
- Buttons:** "Add...", "Edit...", "Remove", and "Add Overflow Location".
- Options:**
 - ☒ Verify images after they are created
 - ☐ Precalculate Progress Statistics
 - ☐ Create directory listings of all files in the image after they are created
- Buttons:** "Start" and "Cancel".

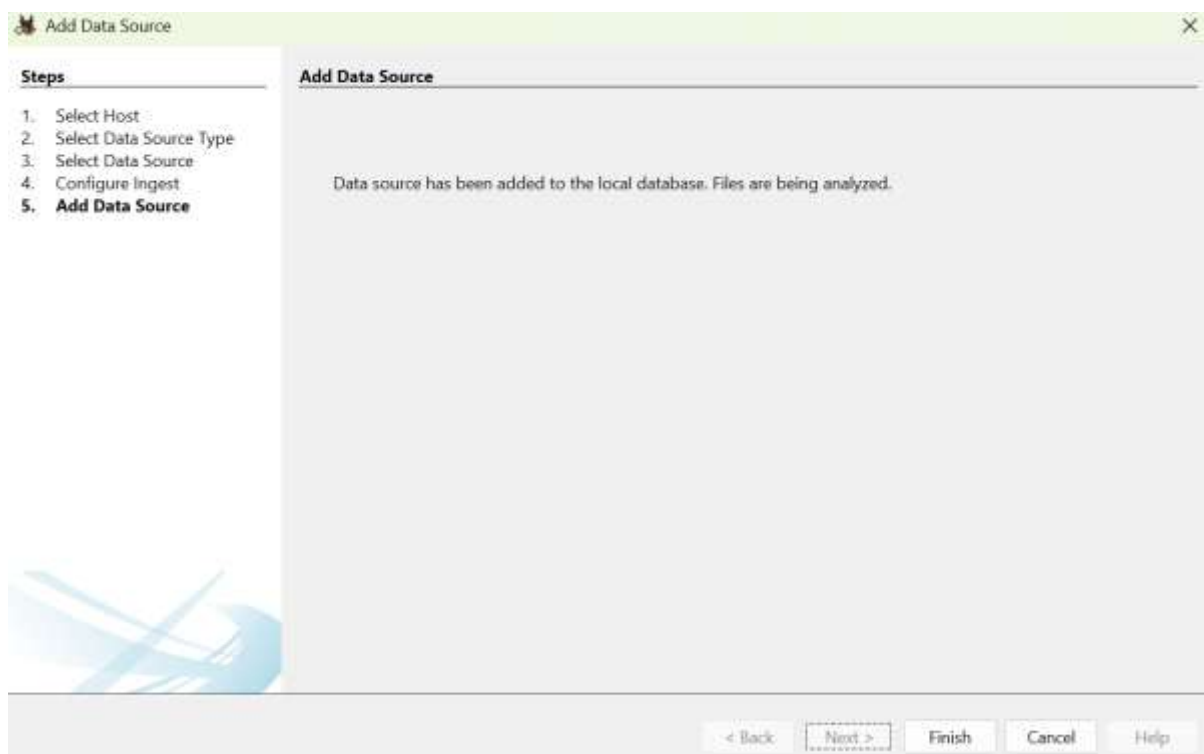
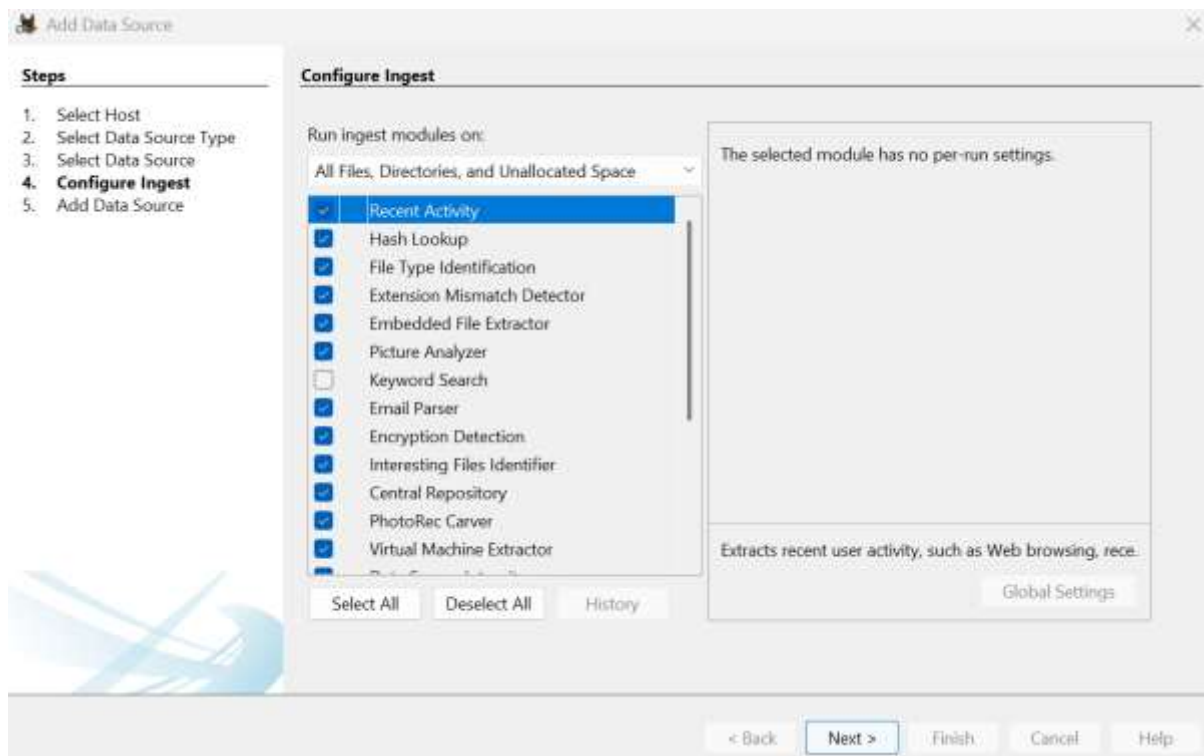
Once acquiring is complete, it will create a text file including all the information it has acquired.





Hash values are matched.





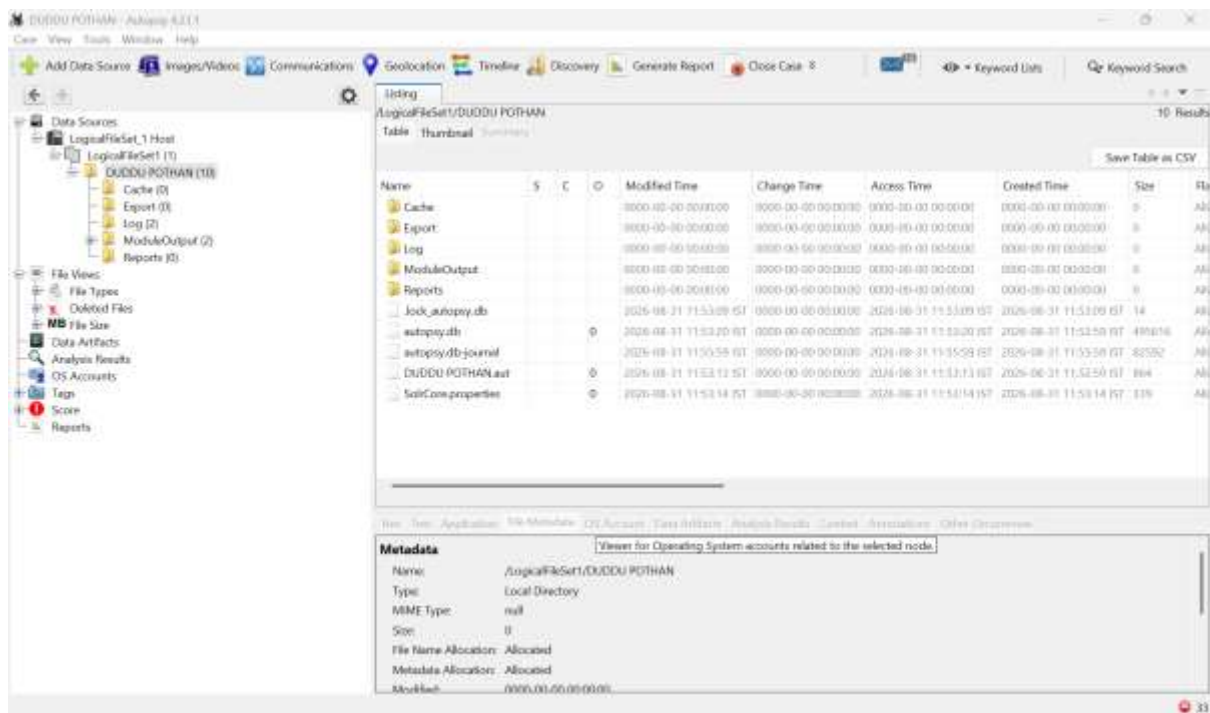
4. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.

- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- **Use the Tree Viewer:**
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**
 - You can perform specific keyword searches using the Keyword Search module.
 - Use pre-configured lists or enter custom keywords.
- **File Analysis:**
 - Navigate through files and folders under the File Types or File System section.
 - Open, view, or export files for further examination.
- **Timeline Analysis:**
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- **Hash Analysis:**
 - Compare file hashes with known databases to identify known good or bad files.



6. Reporting

- **Generate a Report:**
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.

Generate Report

Select and Configure Report Modules

Report Modules:

- ☒ HTML Report
- ☐ Excel Report
- ☐ Files - Text
- ☐ Data Source Summary Report
- ☐ Save Tagged Hashes
- ☐ Extract Unique Words
- ☐ TSK Body File
- ☐ Google Earth KML
- ☐ CASE-UCO
- ☐ Portable Case

A report about results and tagged items in HTML format.

Header:

Footer:

< Back Next > Finish Cancel Help

Report Generation Progress...

Complete

HTML Report : C:\Users\duddu\...Reports\DUDDU POTHAN HTML Report 08-31-2026-11-57-09\report.html

Complete

Cancel Close

Report Navigation

- Case Summary
- Tagged Files (0)
- Tagged Images (0)
- Tagged Results (0)

Autopsy Forensic Report

Case: DUCOU POTIUM

Case Number: 89240041354

Number of Data Sources: 1

Estimate: DUCOU POTIUM

Image Information:

Image Information

Software Information:

Autopsy Version:	4.23.1
Android Analyzer Module:	4.23.1
Android Analyzer (JLEAPP) Module:	4.23.1
Control Repository Module:	4.23.1
DJI Drone Analyzer Module:	4.23.1
Data Source Integrity Module:	4.23.1
Email Parser Module:	4.23.1
Embedded File Extractor Module:	4.23.1
Encryption Detection Module:	4.23.1
Extension Mismatch Detector Module:	4.23.1
File Type Identification Module:	4.23.1
GPX Parser Module:	1.2
Hash Lookup Module:	4.23.1
Interesting Files Identifier Module:	4.23.1
PhotoRec Carver Module:	7.0
Picture Analyzer Module:	4.23.1
Recent Activity Module:	4.23.1
Virtual Machine Extractor Module:	4.23.1
YARA Analyzer Module:	4.23.1
iOS Analyzer (JLEAPP) Module:	4.23.1

Ingest History:

Job 1:

Data Source:	LogicalFileSet1
Status:	COMPLETE
Enabled Modules:	- Recent Activity - Hash Lookup - File Type Identification - Extension Mismatch Detector - Embedded File Extractor - Picture Analyzer - Email Parser - Encryption Detection - Interesting Files Identifier - Control Repository - PhotoRec Carver - Virtual Machine Extractor - Data Source Integrity - Android Analyzer (JLEAPP) - DJI Drone Analyzer - YARA Analyzer - iOS Analyzer (JLEAPP) - GPX Parser - Android Analyzer



Generated by Autopsy Open Source Digital Forensics Platform - www.autopsy.org

7. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.